

12/05/2022 Flow Visualisation Pre-Lab

Introduction

creation of images $\rightarrow$ build a mental picture / model of a flow



chart the motion of a small suspended particle
(follow the path of a fluid element)

computer-assisted PTV (particle tracking velocimetry)

computer-assisted PIV (particle image velocimetry)

introduce of a visualising medium to selectively points of flow

x not precise x might disturb the flow field

✓ using laser to selectively activate the continuously dispersed dye

coating on object

x only shows the effect of the near wall flow

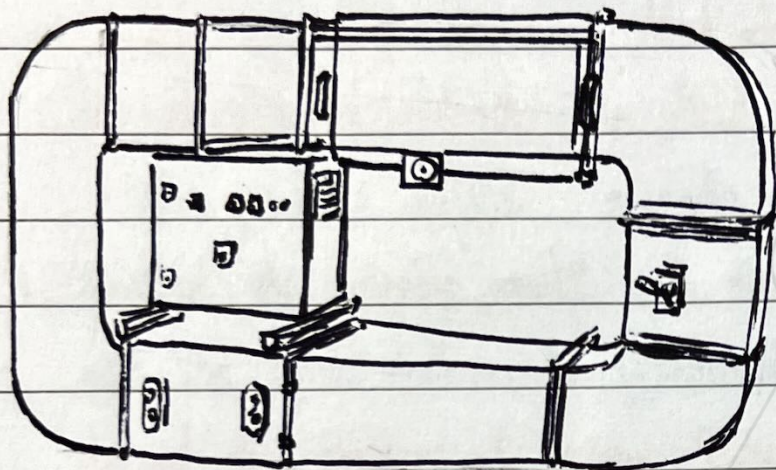
✓ highlight the separation and reattachment points

interferometry 干涉法, Schlieren 射线 and shadowgraph

✓ high speed, thermal and two phase flows

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Experimental apparatus

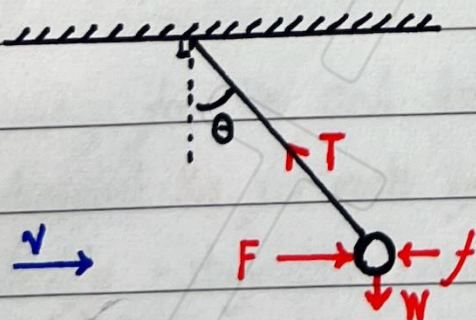


- a circular cylinder model
- a cambered airfoil model
- + mounted on a base plate

- small wind tunnel
- low speed
- powered by a fan
- + vapour generator
- + fog effect 雾化
- + electrically heated wire
- + generate streaklines
- + small foam cylinder
- + estimated the airspeed

Experimental technique

estimation of airspeed



$$\frac{\text{mass } m}{\text{frontal Area } A} \approx 0.3192 \text{ kg/m}^2$$

$$\text{Drag force } f = 0.5 \rho A v^2$$

* represent drag force as D than f

Experimental procedure

1. Mount the circular cylinder and start the flow

estimate the speed and Reynolds number of the flow by θ

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2. Record the significant features of the cylinder flow
Consider separately the near field flow about the cylinder and the characteristics of the far field flow
+ use gridded outline + make approximate measurement
3. Replace the cylinder model with the airfoil model
set the airfoil incidence to approximately $5^\circ/10^\circ/30^\circ$
record the pattern of vortex filaments about the airfoil
4. One additional study of my own choosing
* stall leading by the large angle of attack

Experimental uncertainty

Take particle to comprise spherical drops of water $D = 3-9 \mu\text{m}$
 $F_{\text{drag}} = 3\pi\mu_{\text{air}} D (U_0 - U_p)$ valid for the very low Re

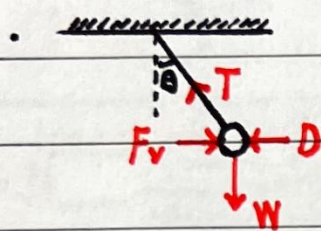
Preparing for the Oral Assessment

- calculated Re for each of my studies
- identify - stagnation point - separation point - Areas of laminar and turbulent flow - Areas of steady and unsteady flow in my sketches
- Map the extent of unsteady region on the sketches
- Indicate the special extent boundary between steady and unsteady flow

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Calculation

- Find μ_a in related temperature T



$$\hookrightarrow T \cos \theta = W = mg$$

$$\hookrightarrow T \sin \theta = D = \frac{1}{2} \rho A v^2$$

$$\therefore \tan \theta = \frac{T \sin \theta}{T \cos \theta} = \left(\frac{mg}{\frac{1}{2} \rho A v^2} \right)^{-1} = \frac{\rho A v^2}{2mg} = (0.1596)^{-1} \frac{\rho v^2}{g}$$

$$\therefore \theta = \arctan \left(6.2657 \frac{\rho v^2}{g} \right)$$

$$v = \sqrt{0.1596 \tan \theta \frac{g}{\rho}} \quad (= \sqrt{\frac{2 \tan \theta g m}{\rho A}})$$

- St Strouhal number = $\frac{f \text{ frequency of vortex shedding behind bodies } L}{v}$

- Re = $\frac{\rho U D}{\mu}$

- $F_D = 3\pi \mu_a D (U_a - U_p) = m a = m \frac{dU_p}{dt}$

$$\therefore \int \frac{1}{U_a - U_p} dU_p = \int \frac{3\pi \mu_a D}{m} dt$$

$$\therefore \int (U_a - U_p)^{-1} dU_p = \left(\frac{3\pi \mu_a D}{m} \right) t + C$$

$$\text{Let } u = U_a - U_p \quad \therefore \frac{du}{dt} = -1 \quad \therefore dU_p = -du$$

$$\therefore \int -u^{-1} du = -\ln|u| + D = -\ln|U_a - U_p| + D = \left(\frac{3\pi \mu_a D}{m} \right) t + C$$

$$\text{When } t=0, U_p=0 \quad \therefore -\ln|U_a| + D = 0 \quad \therefore D = \ln|U_a|, C=0$$

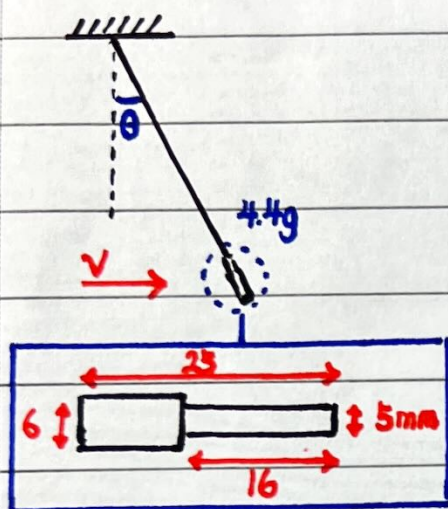
$$\therefore -\ln|U_a - U_p| + \ln|U_a| = \left(\frac{3\pi \mu_a D}{m} \right) t + C = \left(\frac{3\pi \mu_a D}{m} \right) t$$

$$\therefore \ln \left| \frac{U_a}{U_a - U_p} \right| = \left(\frac{3\pi \mu_a D}{m} \right) t \quad \therefore t = \frac{m}{3\pi \mu_a D} \ln \left| \frac{U_a}{U_a - U_p} \right|$$

$$\therefore m = \rho \frac{4}{3} \pi r^3$$

$$\therefore t = \frac{4 \rho \pi r^3}{9 \pi \mu_a D} \ln \left| \frac{U_a}{U_a - U_p} \right| = \frac{4 \rho r^3}{9 \mu_a D} \ln \left| \frac{U_a}{U_a - U_p} \right|$$

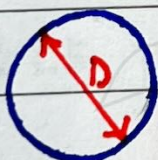
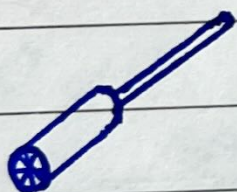
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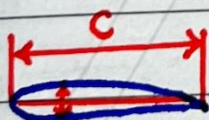
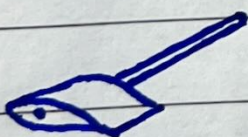
Dial	θ
10	7°
8	5.5°
6	4°
4	2.5°
2	1°
1	0.5°

$$V = \sqrt{\tan \theta \frac{4.81}{1.225} \frac{2 \times 0.0044}{1.34 \times 10^{-4}}} \text{ cm/s}$$

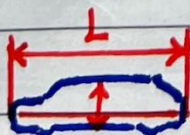
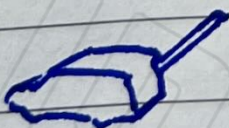
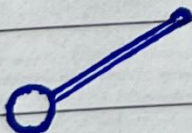
$$T = 16^\circ \text{C} = 289.15 \text{K}$$



$$D = 52 \text{mm} = 0.052 \text{m}$$



$$C = 98 \text{mm} = 0.098 \text{m} \quad \text{thickness} = 0.013 \text{m}$$

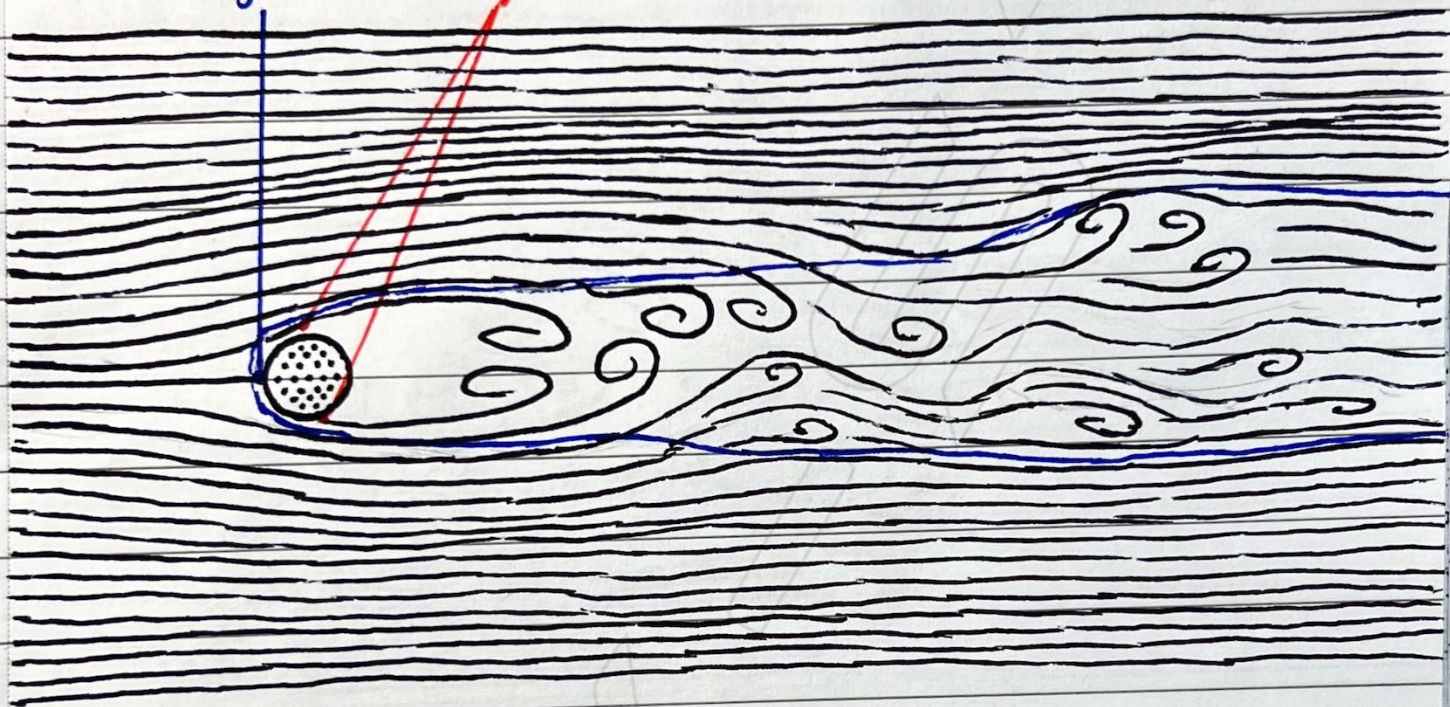


$$L = 16.6 \text{cm} = 0.166 \text{m} \quad \text{thickness} = 0.058 \text{m}$$

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stagnation

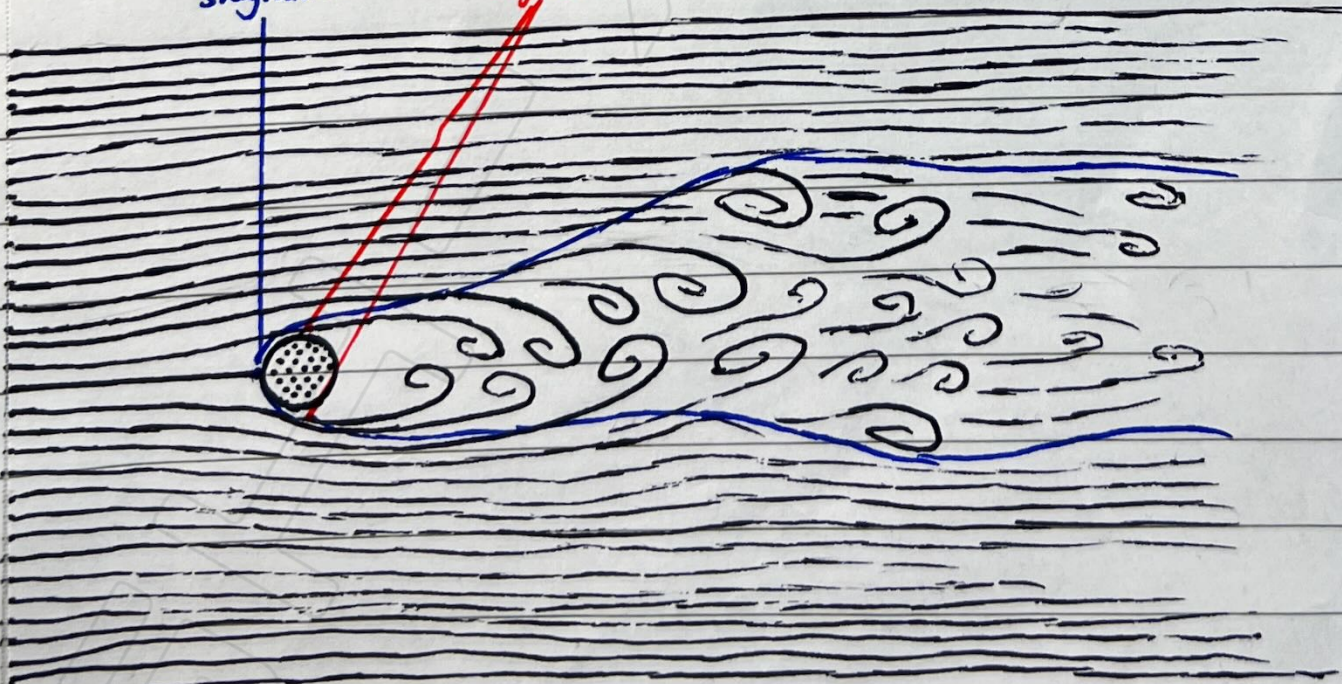
separation



cylinder

stagnation

separation

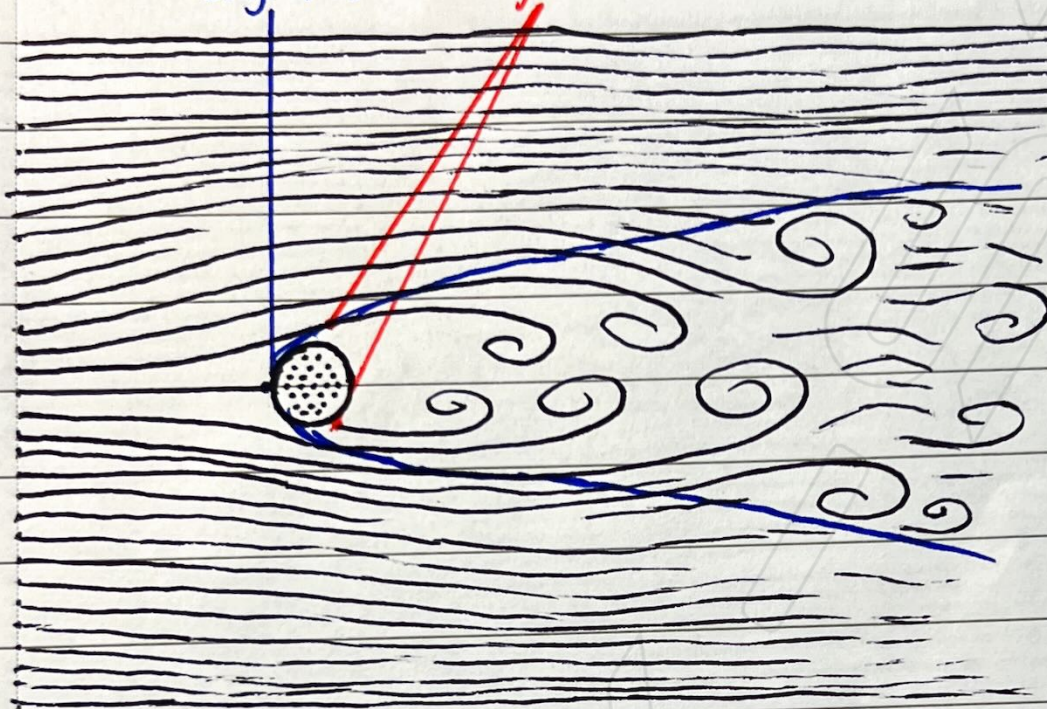


cylinder

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stagnation

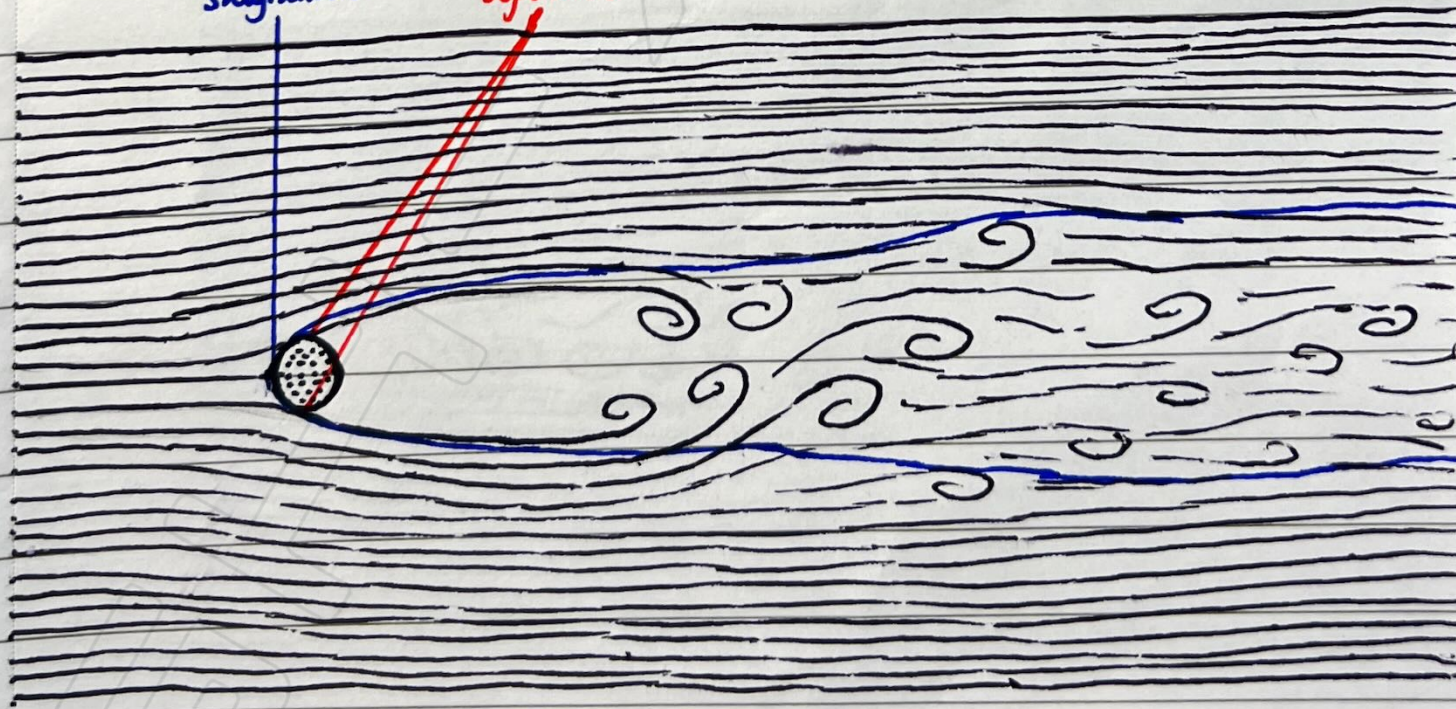
separation



cylinder

stagnation

separation

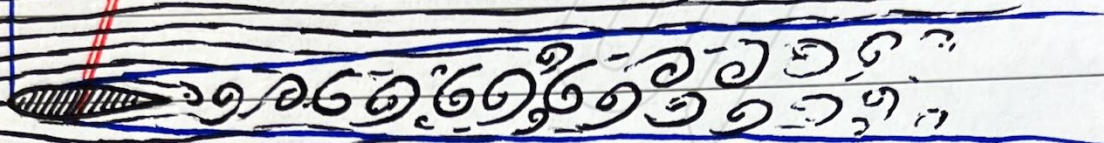


cylinder

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stagnation

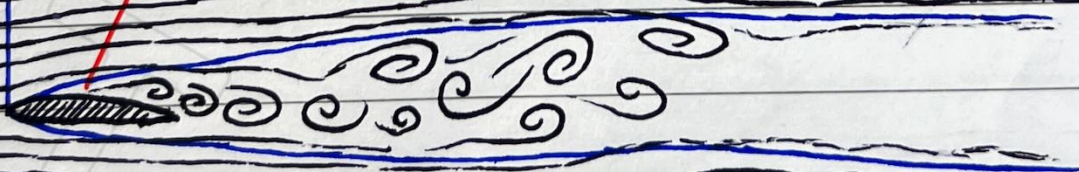
separation



aerofoil

stagnation

separation



aerofoil

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stagnation

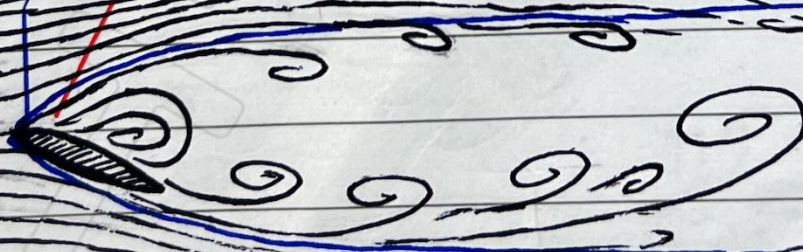
separation



aerofoil

stagnation

separation

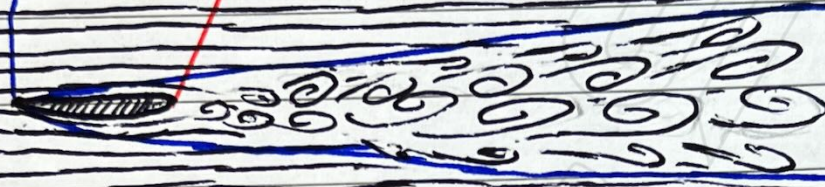


aerofoil

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stagnation

separation



reversed airfoil

stagnation

separation

stagnation

separation

perpendicular airfoil

12/05/2022 Flow Visualisation During Lab

stagnation

separation



O ball

stagnation



car model

12/06/2022 Flow Visualisation Post-Lab

1 How does real cylinder flow compare with inviscid, incompressible flow?



inviscid,
incompressible flow



real cylinder flow

- separation point • turbulent flow
- + skin friction due to viscosity that causes vorticity
- + speed high enough to overcome pressure gradient to separate flow

2 Briefly explain the extent to which the upstream flow is affected by the presence of the body

The upstream flow is affected due to the pressure caused by the fluid not perfectly flowing around the objects. This affects the oncoming flow by deflecting the air. This deflection propagates forwards until the pressure is too low to affect the oncoming air.

The spacing between streamlines decreases as the flow speed increases, since the streamlines closer to the body are most curved and closer together closer to the body, the pressure gradient is higher as the flow has to squeeze through the space around the body.

In experiments, the larger the surface area, the less smooth airflow will move over the object as the separation points move further apart.

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to each other. The distribution of upstream

3 Is the stagnation point at the geometric leading edge in all cases?

Yes, but more downward with the rise of angle of attack

4 Where does separation occur on the models? Are there any locations where flow reattachment can be observed?

Flow reattachment only happened when test the steel ball.

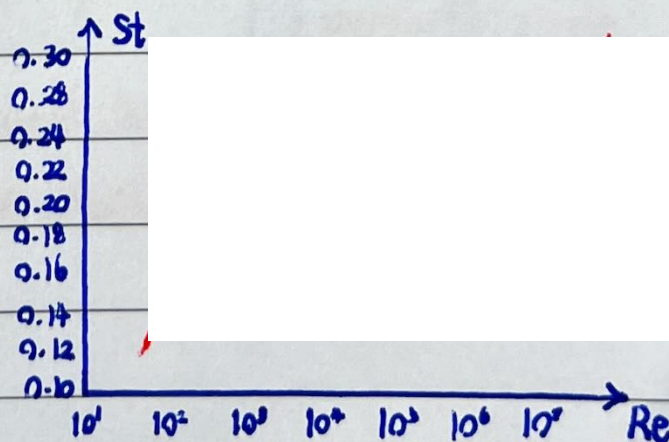
separation always occurs when flow leaving the trailing edge.

5 How does the flow field and St number compare to those suggested by literature for the Re number?

Experiment

Dial	Re	St
1		
1.5		
2		
2.5		

Suggested value



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6 Briefly summarise any sources of experimental error and uncertainty that identified during the experiment

- Small boundary layer between walls of wind tunnel and objects which interact with downstream flow due to proximity
- + Affect particle drag
- + Affect the observation and sketching
- + Affect the frequency of vortex when counting
- Heating by the wire
- Water particles (but very slightly)

time particles need to reach $n\%$ of air speed = t

$$\therefore t = \frac{4\pi r^3}{9\mu_{\text{air}}} \ln \left| \frac{U_a}{U_a - U_p} \right|$$

n	mass of particles (r)	t (s)
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t is really small which could be ignored when particle radius is small, but need to be considered when particle mass become larger.